INTRODUCTION TO COMPUTER NETWORKS

Key Terms

Term	Definition
Computer Network	A set of devices (nodes) connected by communication links to exchange data and share resources.
Node	Any device that can send and/or receive data (computer, printer, server, phone).
Communication Link	Wired or wireless path that carries signals between nodes.
Resource Sharing	Making hardware, software and data available to users irrespective of location.
Client-Server	Clients request services; a server provides services and responses.

2 Marks

Q1. Define a Computer Network. **L1 · 2M**

ANSWER

A **computer network** is a group of computers and other devices linked together by communication links so that they can communicate with one another and share resources, data and applications.

(OR)

A network is a set of devices called **nodes** connected by communication links. A node may be a computer, printer or any device capable of sending and/or receiving data.

Q2. List any four applications of computer networks. **L1 · 2M**

ANSWER

1. **Resource sharing**: Hardware such as printers and scanners, and software or files stored on a server, can be used by many users on the network.
2. **Business communication**: Employees exchange information quickly using email, VoIP calls and desktop sharing for remote work.
3. **E-commerce**: Buying and selling of goods and services is done online through websites and payment systems such as UPI.

4. **Home applications**: Families use networks for Internet access, online entertainment, gaming and sharing files between personal devices.
5. **Mobile applications**: Smartphones support texting, GPS navigation, m-commerce and NFC based short-range services.
6. **Education**: Colleges use networks for ERP, online exams, e-learning and shared laboratory systems.

Q3. Mention any four advantages of computer networks. **L2 · 2M**

ANSWER

1. **Resource sharing**: Programs, printers and data can be accessed by authorized users from any connected system.
2. **High reliability**: Important files can be kept on more than one machine, so work continues even if one system fails.
3. **Cost saving**: Several personal computers with shared servers often cost less than one large mainframe for the same work.
4. **Scalability**: When the workload increases, more computers can be added to improve performance gradually.
5. **Communication medium**: People at different places can work together using email, chat and shared online documents.
6. **Increased productivity**: Many users can enter and process the same shared data at the same time.

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5 Marks

Q4. Explain the concept of a computer network and discuss its applications in various fields. **L2 · 5M**

ANSWER

Concept: A computer network is a collection of interconnected computers and devices that exchange information and share resources.

The devices on a network are called **nodes**. Nodes are joined by **communication links**, which may be wired or wireless. Two computers are said to be interconnected if they can exchange information.

A network may be small, such as a college LAN, or very large, such as the Internet (a network of networks).

Many applications use the **client-server model**. Clients request services and servers provide data or processing. Example: A web browser (client) requesting a page from a web server.

Component	Meaning	Example
Node	Device that sends or receives data	PC, printer, server
Link	Path that carries signals	UTP cable, Wi-Fi
Purpose	Communication and resource sharing	Files, Internet, ERP

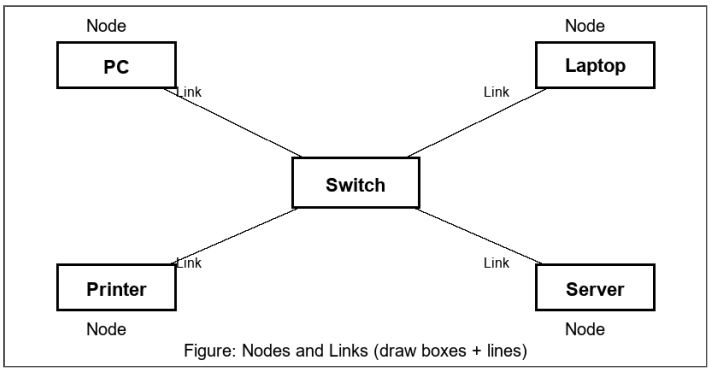


Figure: Nodes and Links

Applications in various fields:

1. **Business**: Printer/file sharing, email, VoIP, desktop sharing and e-commerce with customers and suppliers.
2. **Home**: Internet access, entertainment, gaming and file sharing between personal devices.
3. **Mobile**: Texting, GPS navigation, m-commerce and NFC based short-range services.
4. **Education, healthcare and banking**: College ERP and online exams, hospital systems and telemedicine, ATM, UPI and core banking.

TYPES OF NETWORKS

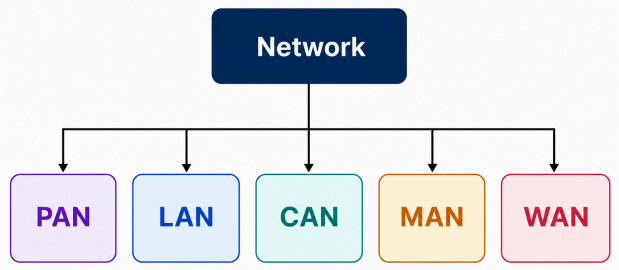


Figure: Types of computer networks (by geographical area)

Key Terms

Term	Full Form	Meaning
PAN	Personal Area Network	Very short-range network around one person and personal devices.
LAN	Local Area Network	Network within a small area such as a room, building or campus.
CAN	Campus Area Network	Network connecting multiple LANs within a college or office campus.
MAN	Metropolitan Area Network	Network covering a city or a large metropolitan area.
WAN	Wide Area Network	Network spanning cities, countries or the whole world.

2 Marks

Q5. Differentiate between LAN and WAN. L2 · 2M

ANSWER

Point	LAN	WAN
Full form	Local Area Network	Wide Area Network
Coverage	Room, building or campus	City to country / worldwide

Speed	Very high (Mbps to Gbps)	Comparatively lower; depends on ISP links
Ownership	Usually owned by one organization	Often shared; uses public / leased links
Example	College computer lab	Internet

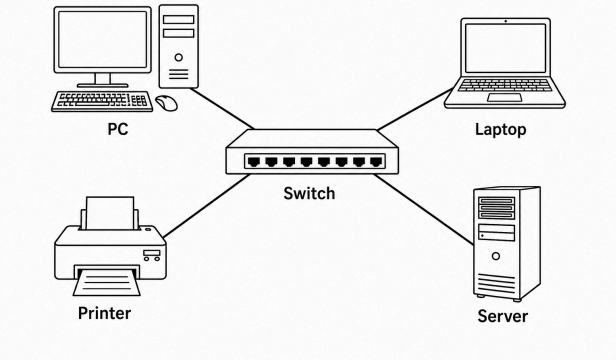


Figure: LAN topology with devices (Star with Switch)

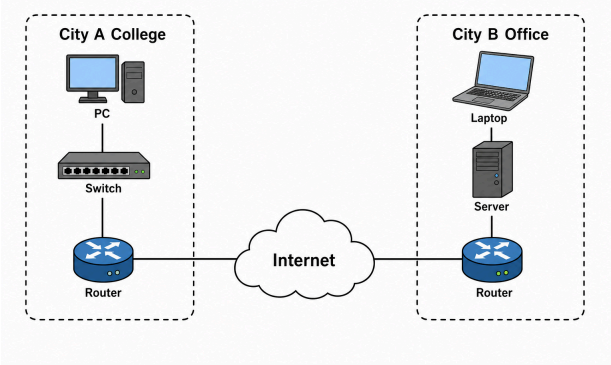


Figure: WAN topology with devices (Sites linked via Internet)

Q6. What is a MAN? Give one example. **L1 · 2M**

ANSWER

A **MAN (Metropolitan Area Network)** is a network that covers a city or a large metropolitan area (about 5 to 50 km). It is larger than a LAN and smaller than a WAN, and is often used to connect several LANs within the same city.

Example: Cable television network in a city, or bank branches across a city linked to one central office.

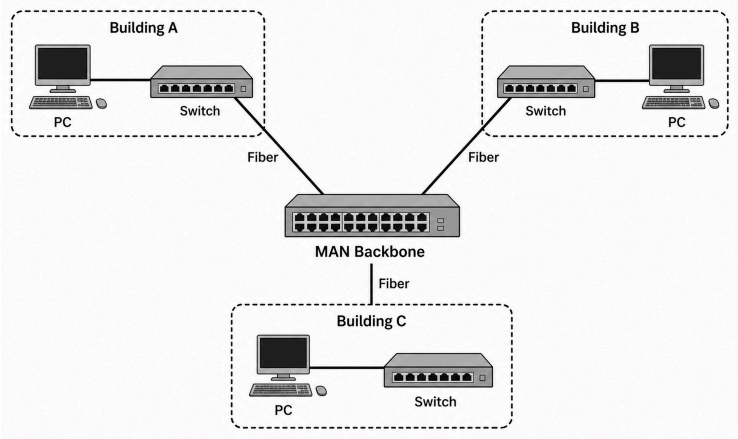


Figure: MAN topology (city buildings linked by fiber backbone)

5 Marks

Q7. Explain the different types of computer networks (PAN, LAN, CAN, MAN, WAN) with suitable examples. L2 · 5M

ANSWER

Computer networks are classified mainly by the **geographical area** they cover. The important types are PAN, LAN, CAN, MAN and WAN.

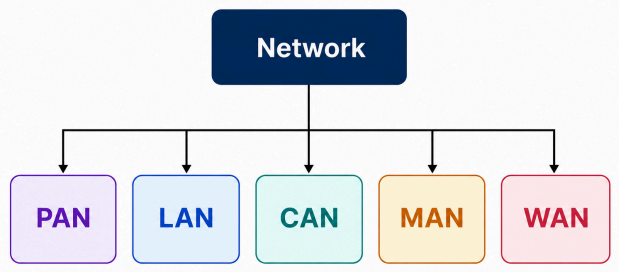


Figure: Classification of computer networks by geographical area

1. PAN (Personal Area Network)

A PAN is a very small network around one person. Devices communicate over a short range (about 1 to 10 metres), often using Bluetooth, NFC or infrared.

Example: A smartphone connected to wireless earbuds, a smartwatch and a laptop.

Figure: PAN topology (wireless links)

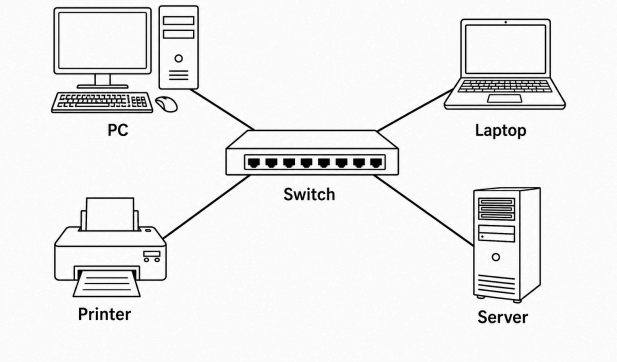


Figure: LAN topology

3. CAN (Campus Area Network)

A CAN connects multiple LANs within a college or corporate campus. It is larger than a single LAN and smaller than a city-wide MAN.

Example: Library, laboratory and admin blocks of a college linked through a campus switch.

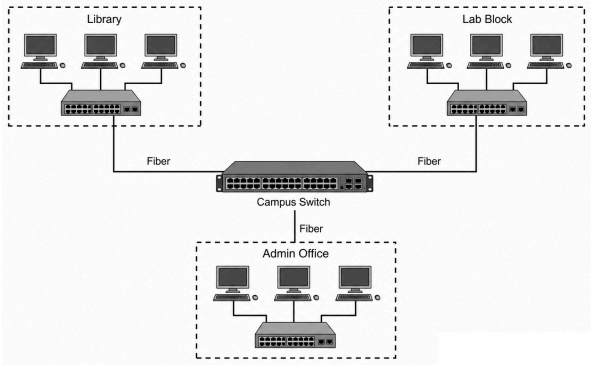
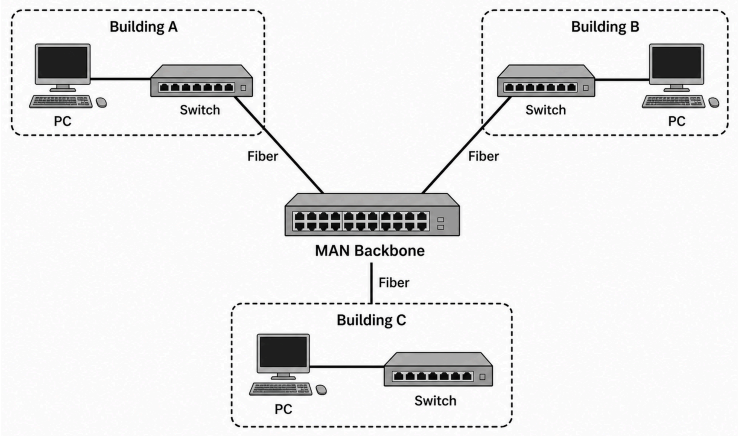


Figure: CAN topology

4. MAN (Metropolitan Area Network)

A MAN covers a city or metropolitan region. It interconnects several LANs that belong to the same organization or service provider across the city.

Example: Bank branches in one city connected to the regional data centre.



5. WAN (Wide Area Network)

A WAN spans a very large area such as a state, country or the whole world. It uses leased lines, optical fibre, microwave or satellite links provided by telecom operators.

Example: The Internet; a company's offices in India and the USA linked through the Internet.

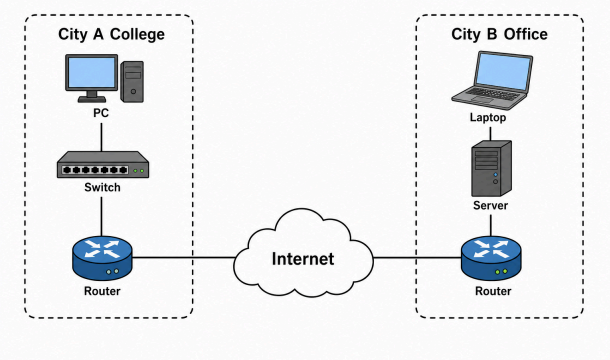


Figure: WAN topology

Q8. Compare LAN, MAN, and WAN based on coverage, speed, ownership, and applications. L3 · 5M

ANSWER

LAN, MAN and WAN differ mainly in how far they spread, how fast they transfer data, who owns them, and where they are used.

Basis	LAN	MAN	WAN
Coverage	Room / building / campus	City (about 5-50 km)	Country / worldwide
Speed	Highest (often Gbps)	High, but usually less than LAN	Lower relative speed; link dependent
Ownership	Single private organization	Private body or service provider	Multiple organizations / public carriers
Applications	File/printer sharing, lab ERP	City offices, cable TV, campus links	Internet, global business, banking
Example	College lab network	City-wide bank network	Internet

NETWORK TOPOLOGIES

Key Terms

Term	Definition
Network Topology	The physical or logical arrangement of nodes and links in a network.
Physical Topology	Actual cable / device layout that can be seen and drawn.
Logical Topology	The path data follows, which may differ from the physical layout.
Terminator	Device at each end of a bus cable that absorbs signals and stops reflection.
Hub / Switch	Central connecting device used in Star Topology.

2 Marks

Q9. Define Network Topology. Name any four types of network topologies. L1 · 2M

ANSWER

Network Topology is the way in which computers (nodes) and communication links are arranged in a network. It describes how devices are interconnected.

Four types: Bus Topology, Star Topology, Ring Topology and Mesh Topology.

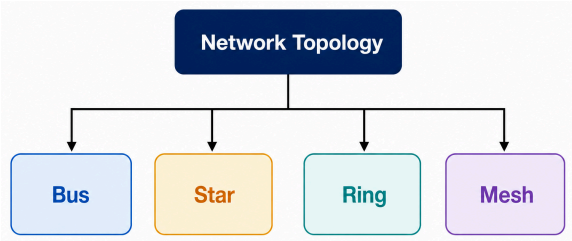


Figure: Types of network topology

(OR)

Common topologies: Bus, Star, Ring, Mesh, Tree and Hybrid.

Q10. State any two advantages of Star Topology. **L2 · 2M**

ANSWER

1. **Easy fault isolation**: If one cable fails, only that node is affected; the rest of the network continues to work.
2. **Easy to add devices**: New nodes can be connected to the central hub or switch without disturbing other stations.
3. **Centralized management**: All nodes connect to a central hub or switch, so monitoring, control and troubleshooting are easier.
4. **Better performance**: Each node has a dedicated link to the central device, which reduces collisions compared to shared-bus topologies.

Q11. List any two disadvantages of Bus Topology. **L2 · 2M**

ANSWER

1. **Single cable failure**: If the main bus cable breaks, the entire network may stop working.
2. **Difficult fault detection**: Finding the exact fault location on a long shared cable is hard.

(OR)

1. **Performance drops with more nodes**: Many stations sharing one cable increase traffic and collisions.
2. **Limited cable length**: The bus has a maximum length; extending it needs repeaters.

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5 Marks

Q12. Explain the different network topologies with neat diagrams. **L2 · 5M**

ANSWER

Network topology is the arrangement of nodes and links. The main topologies are Bus, Star, Ring, Mesh, Tree and Hybrid.

1. Bus Topology

All nodes share one common cable called the bus. Data travels along the bus. Terminators at both ends absorb unused signals.

2. Star Topology

Every node connects to a central hub or switch by its own cable. Communication between two nodes goes through the centre.

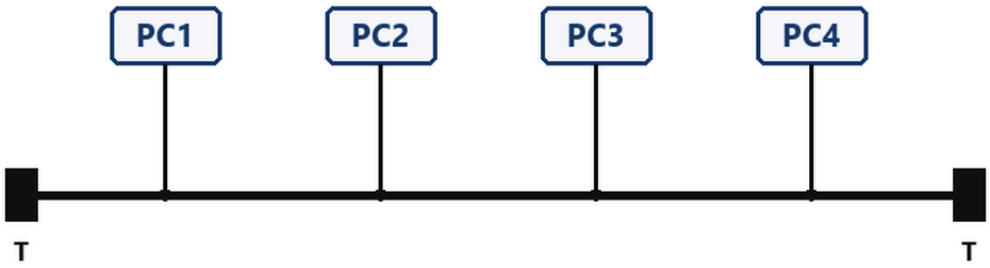

Figure: Bus Topology

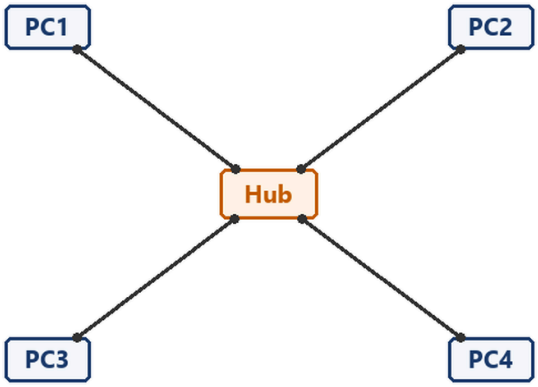

Figure: Star Topology

3. Ring Topology

Each node connects to exactly two neighbours, forming a closed loop. Data usually travels in one direction around the ring until it reaches the destination.

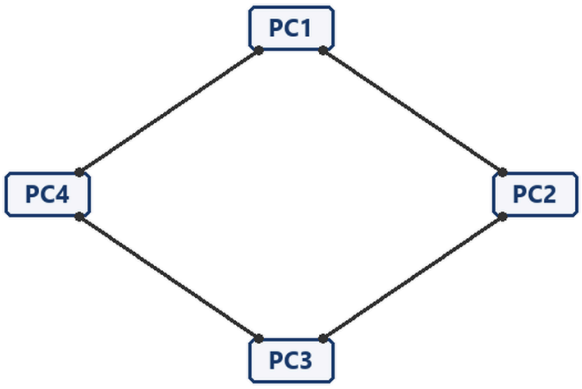

Figure: Ring Topology

4. Mesh Topology

In a full mesh, every node is connected to every other node. This gives many alternate paths and high reliability, but needs many cables.

5. Tree Topology

Nodes are arranged like a tree: a root node branches into intermediate nodes, which connect to leaf nodes. It is a hierarchical combination of star layouts.

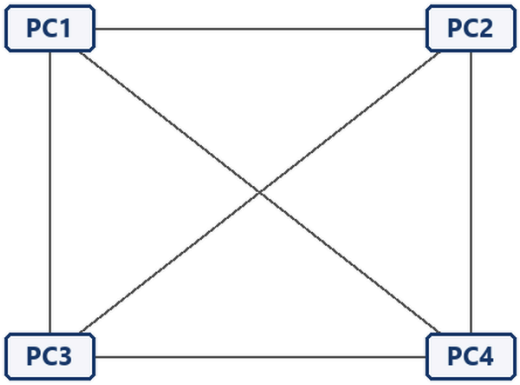

Figure: Mesh Topology

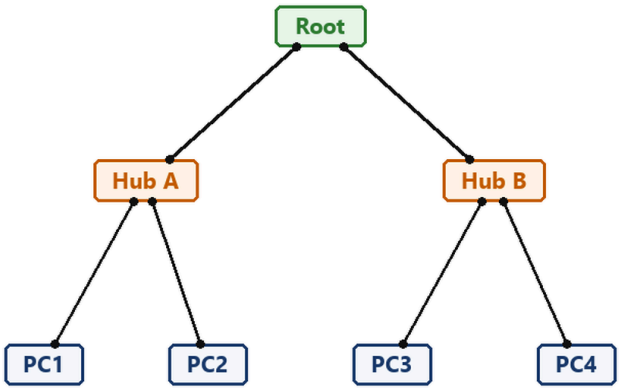

Figure: Tree Topology

6. Hybrid Topology

A hybrid topology combines two or more basic topologies (for example, star networks joined by a backbone). It is used when one topology alone cannot meet all needs.

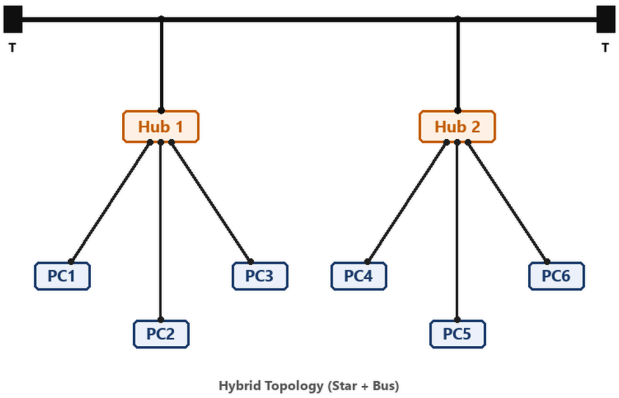

Figure: Hybrid Topology

Q13. Compare Bus, Star, Ring, Mesh, Tree, and Hybrid topologies with their advantages and disadvantages. L4 · 5M

ANSWER

The table below compares the six topologies on structure, main advantage and main disadvantage.

Topology	Structure	Advantage	Disadvantage
Bus	One shared cable; terminators at ends	Simple and cheap to install	Cable break affects whole network
Star	All nodes to central hub/switch	Easy fault finding; easy to expand	Hub/switch failure stops network
Ring	Closed loop; each node has two neighbours	Equal access; orderly data flow	One node/cable fault can break ring
Mesh	Many point-to-point links	High reliability; alternate paths	Costly; complex cabling
Tree	Hierarchical root and branches	Scalable; easy to manage groups	Root failure affects large parts
Hybrid	Mix of two or more topologies	Flexible; can meet mixed needs	Design and maintenance are complex

Quick comparison points:

1. **Cost**: Bus and Star are cheaper; Full Mesh is the costliest.
2. **Reliability**: Mesh is most reliable; Bus and simple Ring are weaker against single faults.
3. **Practical use**: Modern LANs mostly use **Star** (with switches). Large campuses often use **Hybrid** / **Tree** layouts.
4. **Expansion**: Star and Tree are easier to expand than Bus or Ring.

OSI REFERENCE MODEL

Key Terms

Term	Definition
OSI	Open Systems Interconnection: a 7-layer conceptual model for network communication.
ISO	International Organization for Standardization: developed the OSI model.
Layer	A self-contained group of related network functions in the model.
PDU	Protocol Data Unit: the form of data at a given layer (bits, frame, packet, segment, data).
Peer-to-peer	Communication between the same layer on two different systems.

2 Marks

Q14. What is the OSI Reference Model? Why was it developed? **L2 · 2M**

ANSWER

OSI stands for **Open Systems Interconnection**.

The **OSI Reference Model** is a conceptual 7-layer model by **ISO** that describes how data moves between two computers. Each layer has a fixed role, so networks can be designed, taught and troubleshot layer by layer.

Why it was developed: Before OSI, different vendors used incompatible network designs. OSI was developed to provide a **standard framework** for network communication, so that systems from different manufacturers could work together through **interoperability** and **modular design**.

Q15. List the seven layers of OSI Reference Model. **L1 · 2M**

ANSWER

The seven layers of the OSI model (from top to bottom) are:

1. **Application Layer**
2. **Presentation Layer**

3. **Session Layer**
4. **Transport Layer**
5. **Network Layer**
6. **Data Link Layer**
7. **Physical Layer**

Mnemonic (bottom to top): Please Do Not Throw Sausage Pizza Away.

Q16. State the functions of the Transport Layer. **L2 · 2M**

ANSWER

The **Transport Layer (Layer 4)** ensures **end-to-end delivery** of data between source and destination applications. Its main functions are:

- **Segmentation and reassembly**: Breaks large messages into segments and rebuilds them at the receiver.
- **Port addressing**: Uses port numbers to deliver data to the correct application process.
- **Flow and error control**: Controls data rate and retransmits lost segments when needed.
- **Connection management**: Sets up and releases transport connections (TCP) or sends data without connection (UDP).

Q17. What is the role of the Network Layer? **L2 · 2M**

ANSWER

The **Network Layer (Layer 3)** is responsible for **logical addressing** and **routing** of data across different networks. Its role includes:

- **Logical addressing**: Adds source and destination IP addresses to packets.
- **Routing**: Selects the best path for packets from source to destination across interconnected networks.
- **Packet forwarding**: Moves packets from one network to another through routers.
- **Internetworking**: Connects different types of networks into one large network (internet).

5 Marks

Q18. Explain the OSI Reference Model and discuss the functions of each layer. **L2 · 5M**

ANSWER

The **OSI Reference Model** is a 7-layer conceptual model by **ISO** that standardizes network communication. Each layer performs a specific task and passes data to the layer below. Data at each layer is called a **PDU (Protocol Data Unit)**.

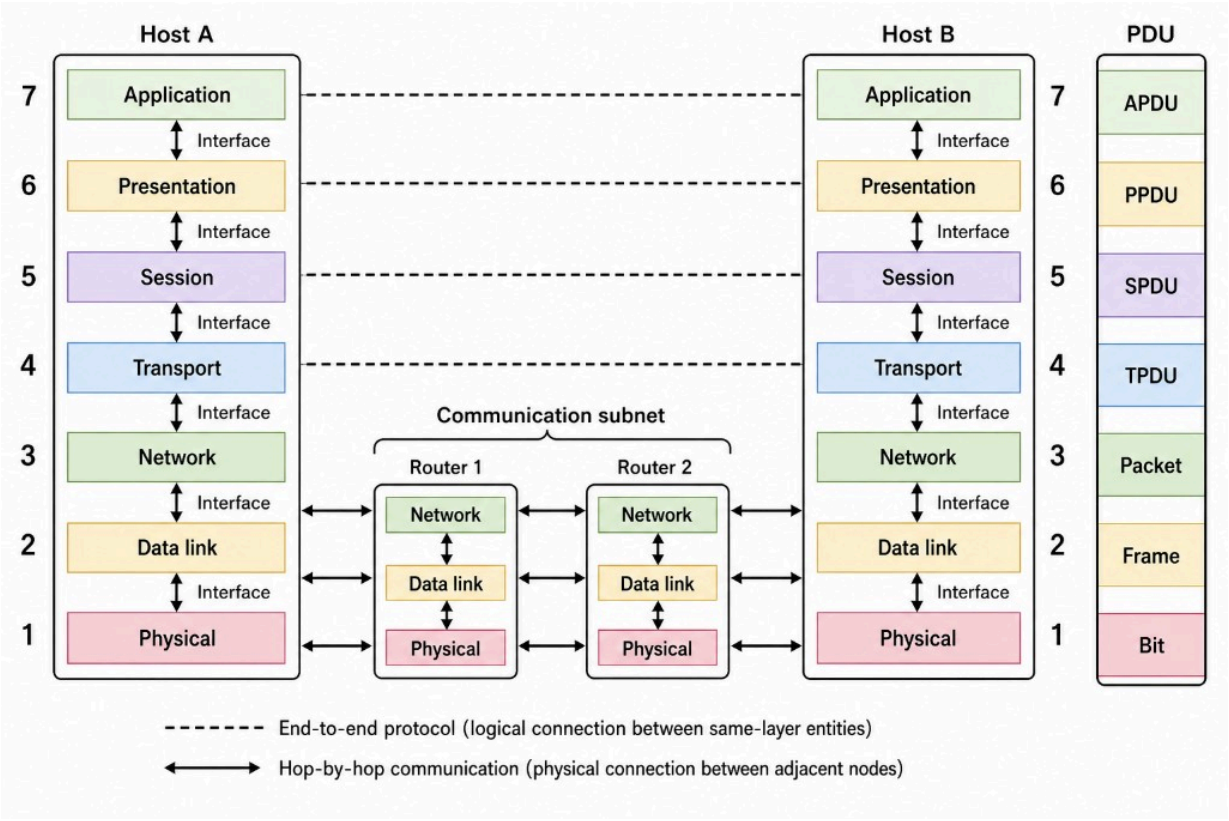

Figure: OSI Reference Model showing hosts, routers, layer-wise communication and PDU

OSI layers at a glance (write in exams):

#	Layer	PDU	Main function	Protocols / devices
7	Application	Data	Provides user interface and network services to applications	HTTP, SMTP, FTP, DNS
6	Presentation	Data	Handles translation, encryption and compression	SSL/TLS, JPEG
5	Session	Data	Establishes, manages and terminates communication sessions	RPC

4	Transport	Segment	Ensures end-to-end delivery with segmentation and flow control	TCP, UDP
3	Network	Packet	Provides logical addressing and routing across networks	IP, Router
2	Data Link	Frame	Creates frames with MAC addressing and error detection	Ethernet, Switch
1	Physical	Bits	Transmits raw bits over the physical medium	Hub, Repeater, Cable

The OSI model has seven layers. Each layer performs a specific function and uses a particular PDU. The functions are explained below from top to bottom.

7. Application Layer (PDU: Data)

This is the topmost layer and the direct interface between the user and the network. It provides network services to user applications such as web browsing, email and file transfer.

- **User interface**: Acts as a window for applications to access network services.
- **Protocols**: HTTP, HTTPS, SMTP, FTP, DNS.

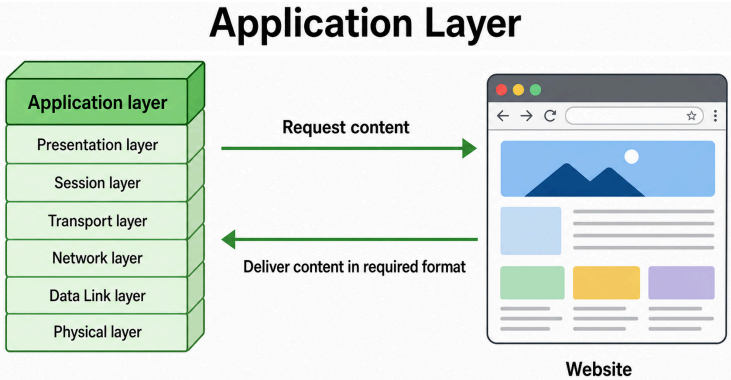

Figure: Application Layer provides network services to user applications

6. Presentation Layer (PDU: Data)

Also called the **translation layer**. It ensures that data is in a form that the receiving application can understand.

- **Translation**: Converts data formats between different systems.
- **Encryption / decryption**: Protects data during transmission (SSL/TLS).

- **Compression**: Reduces data size for faster transfer (JPEG, MPEG).

Figure: Presentation Layer handles encryption, compression and translation

5. Session Layer (PDU: Data)

Acts as a **dialogue manager** between two communicating applications. It establishes, maintains and terminates sessions.

- **Session management**: Opens and closes communication sessions.
- **Dialog control**: Half-duplex or full-duplex communication.
- **Synchronization**: Uses checkpoints so transfer can resume after failure.

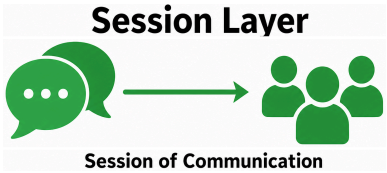

Figure: Session Layer manages sessions between applications

4. Transport Layer (PDU: Segment)

Ensures **end-to-end delivery** of complete messages from source to destination. It uses port numbers to deliver data to the correct application process.

- **Segmentation and reassembly**: Breaks large messages into segments and rebuilds them at the receiver.
- **Flow and error control**: Controls data rate and retransmits lost segments.
- **Protocols**: TCP (reliable), UDP (connectionless).

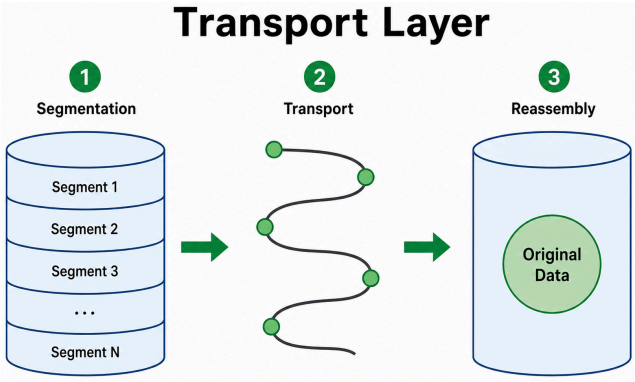

Figure: Transport Layer segments data and ensures end-to-end delivery

3. Network Layer (PDU: Packet)

Responsible for **logical addressing** and **routing** of data across different networks from source to destination.

- **Logical addressing**: Adds source and destination IP addresses to packets.
- **Routing**: Selects the best path across interconnected networks.
- **Devices**: Routers. **Protocols**: IP, ICMP, OSPF.

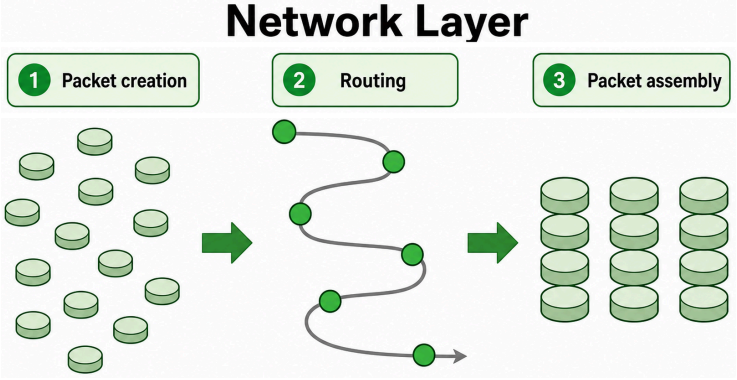

Figure: Network Layer creates packets and routes them across networks

2. Data Link Layer (PDU: Frame)

Ensures reliable **node-to-node delivery** over a single link. It converts packets from the Network Layer into frames.

- **Framing**: Groups bits into frames with start and stop markers.
- **Physical addressing**: Uses MAC addresses in the frame header.
- **Error detection**: Detects damaged frames using CRC. Also handles flow and access control.
- **Devices**: Switches and bridges. **Protocols**: Ethernet, PPP.

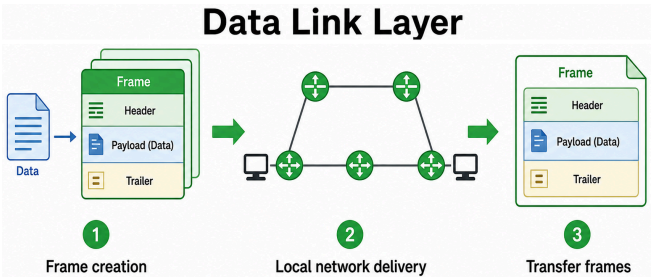

Figure: Data Link Layer creates frames and delivers them on the local network

1. Physical Layer (PDU: Bits)

The lowest layer. It transmits **raw bits** as electrical, optical or radio signals over the physical medium.

- **Bit transmission**: Converts frames into bit streams and sends them over cable, fibre or Wi-Fi.

- **Bit synchronization**: Keeps sender and receiver clocks aligned.
- **Bit rate and topology**: Defines speed and physical layout of the network.
- **Devices**: Hubs, repeaters, modems and cables.

PDU mnemonic (Layer 1 to 7): Bits, Frame, Packet, Segment, Data, Data, Data.

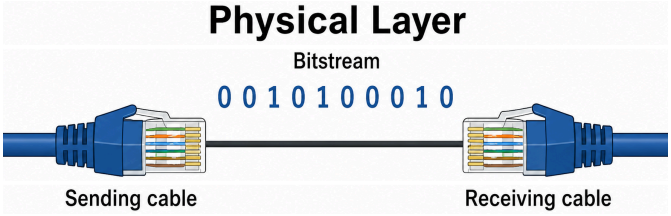

Figure: Physical Layer transmits raw bits over the physical medium

Q19. Describe the services and importance of the OSI Reference Model in data communication. L2 · 5M

ANSWER

The **OSI (Open Systems Interconnection) Reference Model** is a 7-layer conceptual framework developed by **ISO**. It explains how data travels between computers in a structured way. In data communication, it acts as a **blueprint** that describes what services each layer should provide so different systems can communicate effectively.

Services of the OSI Model:

1. **Layered structure**: Network communication is divided into seven layers. Each layer has a clear responsibility and provides services only to the layer directly above it.
2. **Service to upper layers**: Lower layers offer services such as bit transmission, error detection, routing and reliable delivery. Upper layers offer session control, data translation and application support.
3. **Peer-to-peer communication**: The same layer on the sender and receiver works together. For example, the Transport layer at the source communicates with the Transport layer at the destination.
4. **Encapsulation and de-encapsulation**: While sending, each layer adds its own header to the data. While receiving, each layer removes its header and passes the remaining data upward.
5. **Data flow control**: At the sender, data moves down from Application to Physical layer. At the receiver, data moves up from Physical to Application layer.

Layer group	Layers	Main service
Upper layers	Application, Presentation, Session	User services, data format, session management
Lower layers	Transport, Network, Data Link, Physical	End-to-end delivery, routing, framing, bit transmission

Importance in data communication:

1. **Standardization**: It gives a common language for network design. Different manufacturers can build devices that work together.
2. **Modularity**: One layer can be updated or replaced without redesigning the entire network system.
3. **Interoperability**: Heterogeneous systems (different hardware, software and protocols) can exchange data more easily.
4. **Easy understanding**: Students and engineers can study complex networking step by step using the seven layers.
5. **Troubleshooting**: Network problems can be isolated to a specific layer, which saves time in fault finding and maintenance.
6. **Protocol development**: New protocols can be designed for individual layers without disturbing other layers.

TCP/IP REFERENCE MODEL

Key Terms

Term	Definition
TCP/IP	Transmission Control Protocol / Internet Protocol: a practical 4-layer framework and protocol suite used on the Internet.
Network Access Layer	Also called Host-to-Network / Link Layer: handles framing, MAC addressing and bit transmission on the local medium.
IP	Internet Protocol: logical addressing, packaging and routing of packets across networks.
TCP	Reliable, connection-oriented transport protocol (handshake, sequencing, ACK, retransmission).
UDP	Fast, connectionless transport protocol with low overhead and no delivery guarantee.

2 Marks

Q20. What is the TCP/IP Reference Model? Name its layers. L1 · 2M

ANSWER

TCP/IP stands for **Transmission Control Protocol / Internet Protocol**. The **TCP/IP Reference Model** is a layered networking framework that explains how data is communicated between devices using standardized protocols. It is simpler and more practical than the 7-layer OSI model and is the core framework of the modern Internet.

The TCP/IP model has **four layers** (from top to bottom):

1. **Application Layer**
2. **Transport Layer**
3. **Internet Layer**
4. **Network Access (Link) Layer** (also called Host-to-Network Layer)

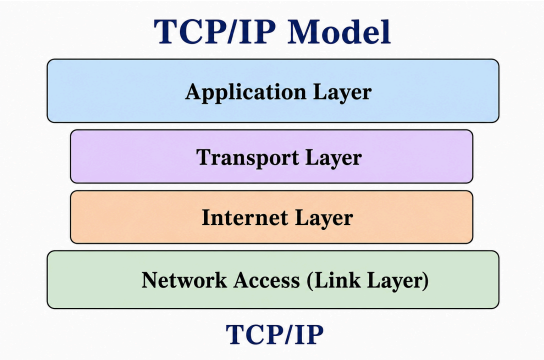

Figure: Layers of the TCP/IP model

5 Marks

Q21. Explain the TCP/IP Reference Model and describe the functions of its layers. **L2 · 5M**

ANSWER

The **TCP/IP Reference Model** is a 4-layer practical framework for network communication. Data moves down the layers at the sender (encapsulation) and up the layers at the receiver (de-encapsulation).

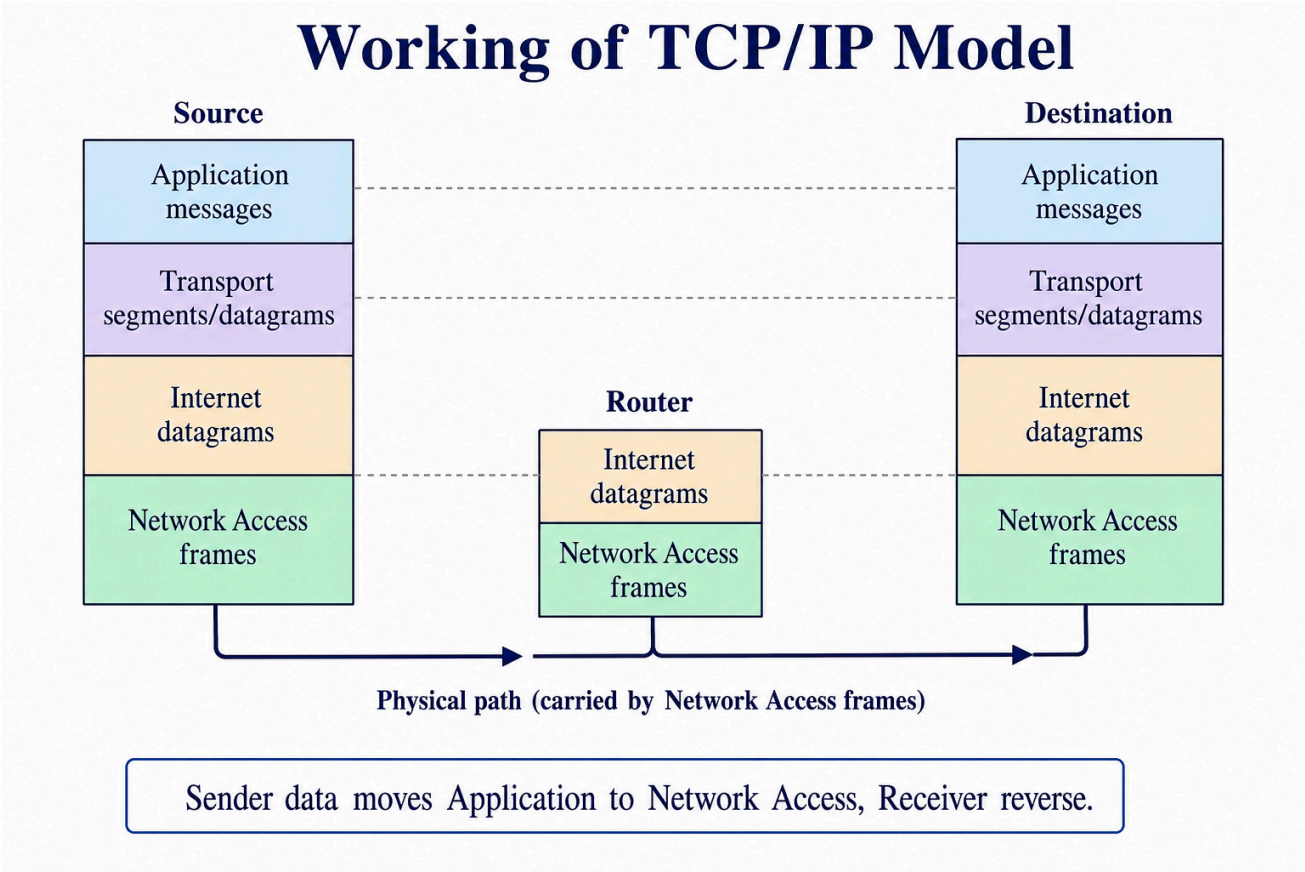

Figure: Working of TCP/IP showing PDUs and router path

- **Application**: Creates user data (messages).
- **Transport**: Adds ports and forms segments (TCP) or user datagrams (UDP).
- **Internet**: Adds IP addresses and forms datagrams for routing.
- **Network Access**: Forms frames and transmits bits on the medium.

TCP/IP layers at a glance:

#	Layer	Main function	Protocols / PDU
4	Application	User services, formatting, sessions	HTTP, FTP, SMTP, DNS / Message
3	Transport	End-to-end delivery and ports	TCP, UDP / Segment or Datagram
2	Internet	Addressing and routing	IP, ICMP, ARP / Datagram
1	Network Access	Framing and bit transmission	Ethernet, Wi-Fi / Frame

The four TCP/IP layers and their functions are explained below from top to bottom.

4. Application Layer

This is the top layer where applications such as browsers, email clients and file-sharing tools interact with the network. It acts as a bridge between user software and lower layers.

- **Services**: Supports protocols such as HTTP, FTP, SMTP and DNS.
- **Extra roles**: May handle data formatting, encryption and session tracking inside applications.

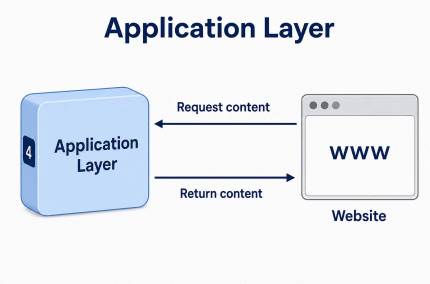

Figure: Application Layer provides services to user applications

3. Transport Layer

Ensures efficient delivery between processes. It breaks large messages into packets, uses **port numbers** so many applications can share the network, and may control flow so the receiver is not overwhelmed.

- **TCP**: Connection setup, error checking, retransmission and correct order.
- **UDP**: Connectionless, low-latency delivery without recovery of lost data.

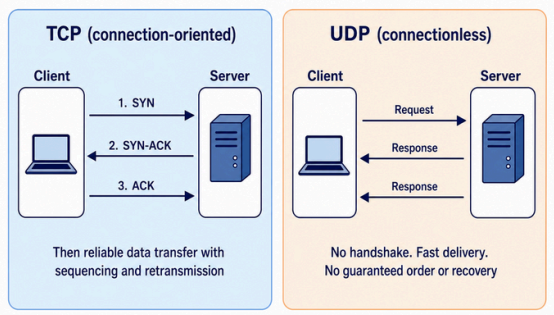

Figure: Transport Layer protocols TCP (handshake) and UDP (direct exchange)

2. Internet Layer

Responsible for moving packets across different networks to the correct destination device.

- Assigns IP addresses to source and destination.
- Selects the best path through routers (routing and path selection).
- Uses IP as the main protocol, with ICMP and ARP for support.

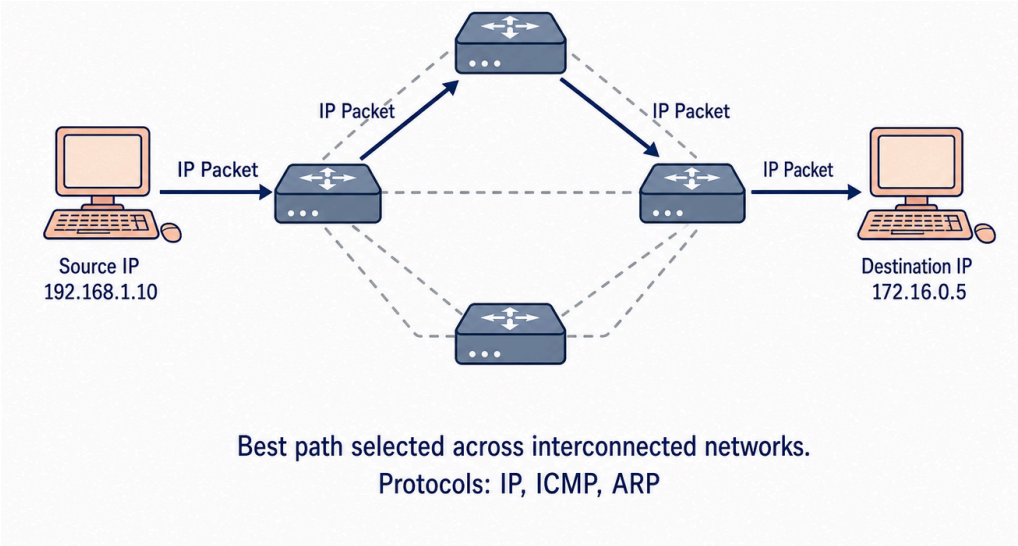

Figure: Internet Layer performs routing and path selection

1. Network Access (Link) Layer

Handles physical transmission over cables, switches and wireless links. It formats data for the medium and delivers it to the next device on the local path.

- **Framing**: Organizes data into frames for transmission.
- **Physical addressing**: Uses MAC addresses within the same network segment.
- **Bit transmission**: Sends and receives raw bits over Ethernet, fibre or Wi-Fi.
- **Error detection**: Uses checksums or CRC on the link.

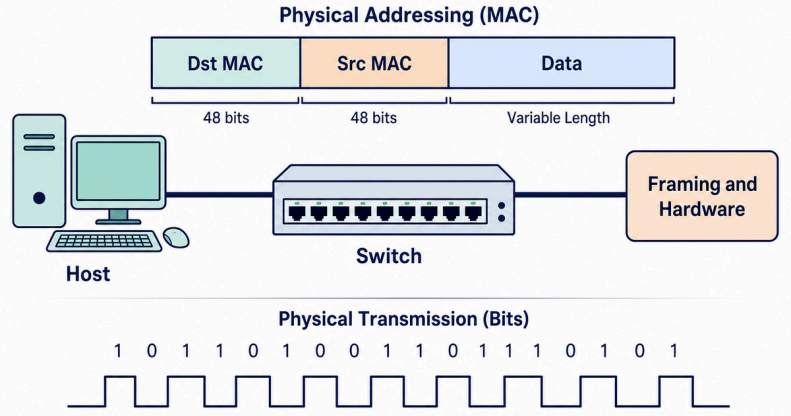

Figure: Network Access Layer handles framing, MAC and bit transmission

OSI vs TCP/IP REFERENCE MODELS

Key Terms

Term	Definition
OSI Model	7-layer theoretical reference model developed by ISO for teaching and designing networks.
TCP/IP Model	4-layer practical protocol suite used on the Internet (also called the Internet model).
Layer mapping	How OSI layers group into TCP/IP layers (for example OSI 5-7 map to TCP/IP Application).
Reference model	A framework that describes roles of layers. It may or may not match a real protocol suite exactly.
Protocol suite	A working set of protocols (such as TCP, UDP, IP, HTTP) used together in real networks.

2 Marks

Q22. Mention any two differences between the OSI and TCP/IP models. **L2 · 2M**

ANSWER

1. **Number of layers**: OSI has **7 layers**, while TCP/IP has **4 layers**.
2. **Nature of model**: OSI is mainly a **theoretical / teaching** reference model. TCP/IP is a **practical** model used on the real Internet.

(Other valid points): OSI was developed by ISO. TCP/IP grew from ARPANET / DoD work. OSI has separate Session and Presentation layers. In TCP/IP those roles are handled inside the Application Layer.

5 Marks

Q23. Compare the OSI and TCP/IP reference models with suitable diagrams and a comparison table. L4 · 5M

ANSWER

Both OSI and TCP/IP explain network communication using layers. OSI is a 7-layer conceptual model. TCP/IP is a 4-layer practical suite used by the Internet. Their layers can be mapped as shown below.

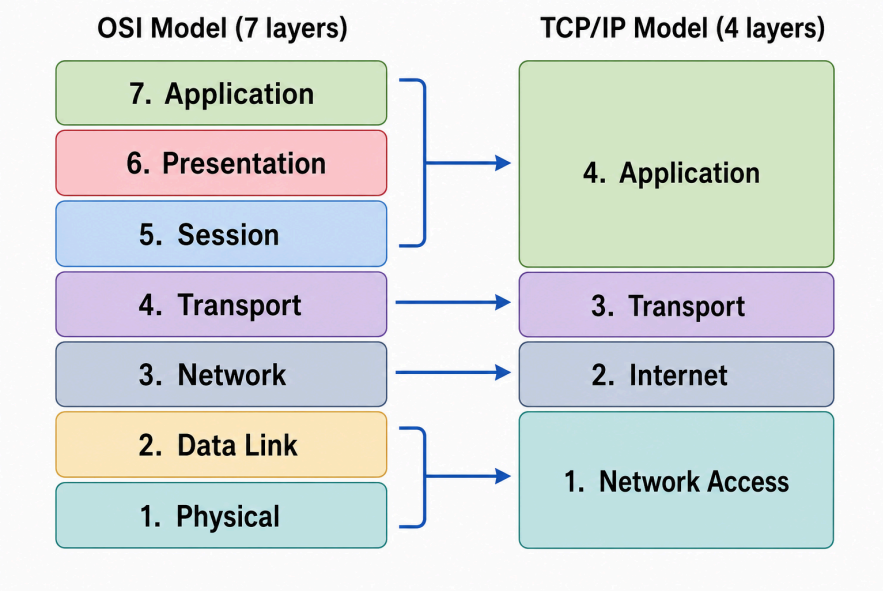

Figure: Layer mapping between OSI (7 layers) and TCP/IP (4 layers)

Layer mapping (OSI to TCP/IP):

- OSI Application + Presentation + Session → TCP/IP **Application Layer**
- OSI Transport → TCP/IP **Transport Layer**
- OSI Network → TCP/IP **Internet Layer**
- OSI Data Link + Physical → TCP/IP **Network Access Layer**

Comparison table:

Point	OSI Model	TCP/IP Model
Number of layers	7 layers	4 layers
Type	Theoretical / reference model	Practical / protocol-driven model
Developer	ISO	DoD / ARPANET community
Session and Presentation	Separate layers (5 and 6)	Combined into Application Layer

Point	OSI Model	TCP/IP Model
Network + Link + Physical	Three separate lower layers	Internet + Network Access (two layers)
Protocols	Model first, protocols later	Protocols first, model describes them
Usage today	Mainly teaching and design	Used on the real Internet
Connection services	Transport and Network can be connection-oriented or connectionless	Transport: TCP/UDP. Internet: mainly connectionless IP

Similarities:

- Both are **layered** models for network communication.
- Both have an Application-related top part and a Transport layer for end-to-end delivery.
- Both support packet-based communication across networks.
- Both help in understanding encapsulation and troubleshooting by layer.

Differences (summary):

- OSI has more layers and clearer separation of Session/Presentation roles.
- TCP/IP is simpler and matches real Internet protocols (IP, TCP, UDP, HTTP, and others).
- OSI is preferred for teaching. TCP/IP is preferred for implementation.

PHYSICAL LAYER & TRANSMISSION MEDIA

Key Terms

Term	Definition
Physical Layer	OSI Layer 1: transmits raw bits as electrical, optical or radio signals over the medium.
Transmission media	The path used to carry signals between devices (wired or wireless).
Guided media	Wired media where signals travel inside a physical path (cable or fibre).
Unguided media	Wireless media where signals travel through air or free space.
Bit rate	Number of bits transmitted per second over the physical medium.

2 Marks

Q24. What are the functions of the Physical Layer? **L2 · 2M**

ANSWER

The **Physical Layer (Layer 1)** is responsible for the actual transmission of bits over the communication medium. Its main functions are:

- **Bit transmission**: Converts frames into a stream of bits and sends them as signals.
- **Signal encoding**: Defines how bits are represented as electrical, optical or radio signals.
- **Bit synchronization**: Keeps sender and receiver clocks aligned for correct sampling.
- **Physical characteristics**: Defines connectors, voltage levels, cable types, topology and data rate.

Q25. Differentiate between Guided Media and Unguided Media. **L2 · 2M**

ANSWER

Guided media (also called wired or bounded media) are transmission paths where signals travel inside a physical cable or fibre, such as twisted pair, coaxial cable and optical fibre.

Unguided media (also called wireless or unbounded media) are transmission paths where signals travel through air or free space as electromagnetic waves, such as radio waves, microwaves and infrared.

Point	Guided Media	Unguided Media
Also called	Wired / bounded media	Wireless / unbounded media
Path	Signals travel inside a physical cable or fibre	Signals travel through air or free space
Examples	Twisted pair, Coaxial cable, Optical fibre	Radio waves, Microwaves, Infrared
Security	Generally more secure (harder to tap)	More open to interception and interference
Mobility	Fixed installation, less flexible	Supports mobile and remote communication

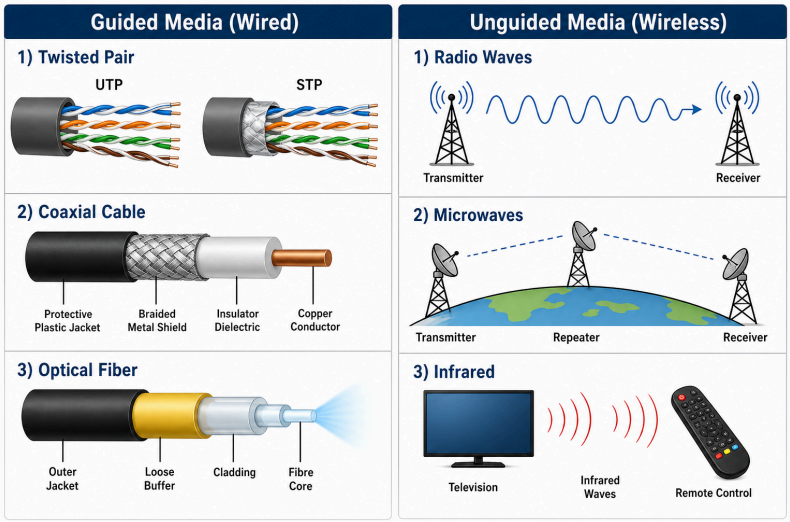

Figure: Guided (wired) vs Unguided (wireless) transmission media

5 Marks

Q26. Explain the functions and characteristics of the Physical Layer. **L2 · 5M**

ANSWER

The **Physical Layer** is the lowest layer of the OSI model. It deals with the hardware means of sending and receiving data as a stream of bits over a transmission medium.

Functions of the Physical Layer:

1. **Bit transmission**: Accepts frames from the Data Link Layer and transmits them as bits over the medium.
2. **Signal representation**: Decides how 0s and 1s are encoded as voltages, light pulses or radio waves.
3. **Synchronization**: Maintains timing so the receiver can correctly identify bit boundaries.
4. **Data rate control**: Defines how many bits can be sent per second (bit rate / bandwidth limits).
5. **Physical topology support**: Works with bus, star, ring and other physical layouts of links and devices.

Characteristics of the Physical Layer:

- **Hardware oriented**: Concerned with cables, connectors, NICs, hubs, repeaters, modems and wireless radios.
- **No addressing / routing**: It does not understand MAC or IP addresses. It only moves bits.
- **Medium dependent**: Rules change for copper, fibre and wireless links.
- **Error unchecked at this layer**: Error detection and framing are handled mainly by the Data Link Layer above it.
- **Defines interface standards**: Examples include Ethernet physical specs, RS-232 and DSL line coding.

Aspect	Physical Layer detail
OSI layer number	Layer 1 (lowest layer)
PDU	Bits
Main job	Move raw bits over the physical medium
Devices	Hub, repeater, modem, cables, antennas
Media types	Guided (wired) and Unguided (wireless)

GUIDED TRANSMISSION MEDIA

Key Terms

Term	Definition
Guided media	Wired media where signals travel inside a physical cable or fibre (bounded path).
UTP	Unshielded Twisted Pair: twisted copper pairs without metallic shielding.
STP	Shielded Twisted Pair: twisted pairs with foil or braid shield against interference.
Coaxial cable	Cable with copper core, insulator, metallic shield and outer jacket.
Optical fibre	Cable that carries data as light through a glass or plastic core using total internal reflection.

2 Marks

Q27. Compare UTP and STP cables. L2 · 2M

ANSWER

Both UTP and STP are twisted-pair cables. The main difference is shielding.

Point	UTP	STP
Full form	Unshielded Twisted Pair	Shielded Twisted Pair
Shielding	No metallic shield	Foil or braid shield present
Interference	More affected by EMI / crosstalk	Better protection against EMI / crosstalk
Cost / weight	Cheaper, lighter, easier to install	Costlier, bulkier, harder to install
Use	Common LANs and telephone lines	Noisy industrial / high-interference areas

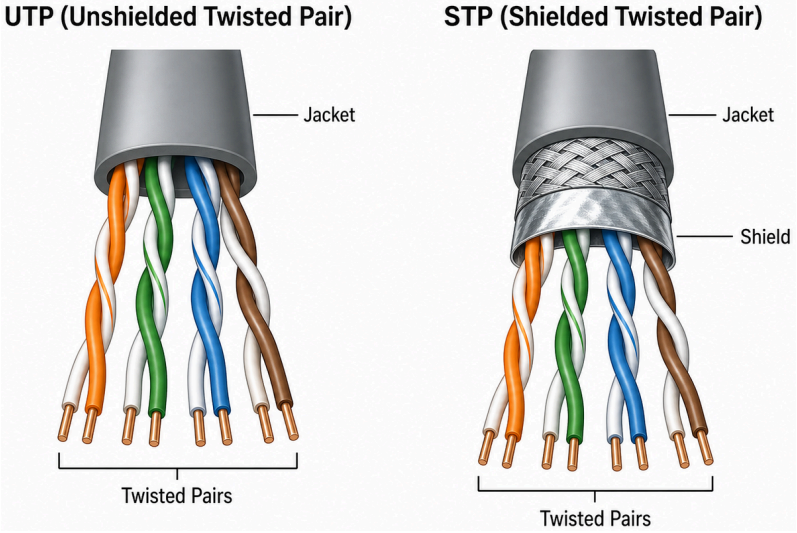

Figure: UTP (no shield) vs STP (with metallic shield)

Q28. State any two advantages of Optical Fiber Cable. **L2 · 2M**

ANSWER

1. **Very high bandwidth**: Optical fibre can carry huge amounts of data at very high speed over long distances.
2. **Immune to EMI**: Because it uses light, fibre is not affected by electromagnetic interference and electrical noise.

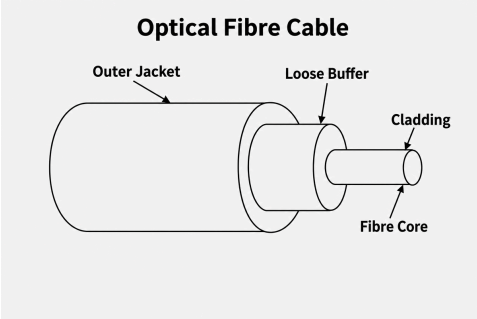

Figure: Structure of optical fibre cable

5 Marks

Q29. Explain the different types of guided transmission media with neat diagrams. **L2 · 5M**

ANSWER

Guided media (also called wired or bounded media) transmit data signals through a physical cable along a fixed path. The three main types are Twisted Pair Cable, Coaxial Cable and Optical Fiber Cable.

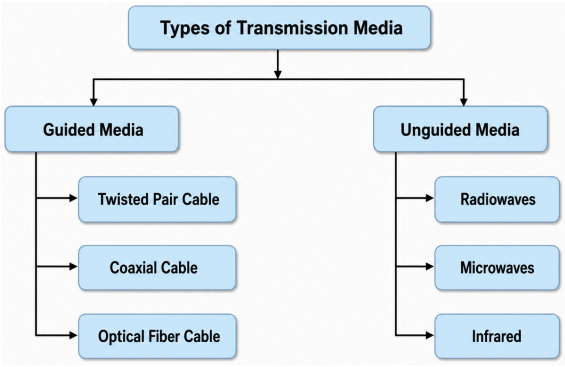

Figure: Classification of transmission media (focus: guided types)

1. Twisted Pair Cable

Insulated copper conductors twisted in pairs to reduce EMI and crosstalk. Cheap, flexible and widely used in LANs and telephone systems.

(a) UTP (Unshielded Twisted Pair): Twisted pairs with no extra metallic shield. Least expensive; easy to install; used in Ethernet LANs and telephone networks. More sensitive to noise than STP.

(b) STP (Shielded Twisted Pair): Twisted pairs with foil or braided metallic shield. Better EMI protection for noisy environments, but costlier, bulkier and harder to install than UTP.

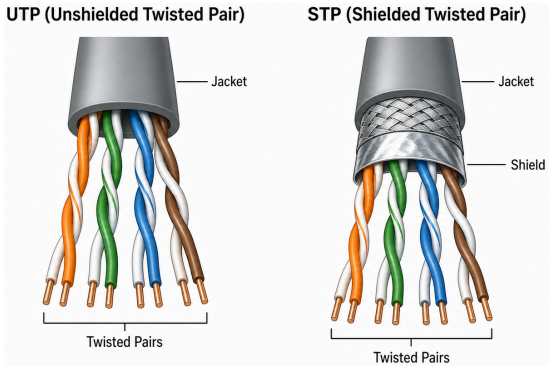

Figure: UTP (no shield) and STP (with metallic shield)

2. Coaxial Cable

Central copper conductor + dielectric insulator + metallic shield + outer jacket. Higher bandwidth than twisted pair; used in baseband or broadband mode.

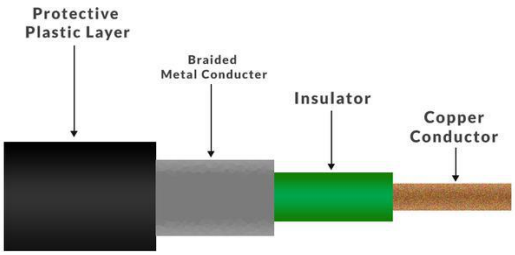

Figure: Structure of coaxial cable

- **Advantages:** Higher bandwidth than twisted pair; better signal quality; less affected by EMI.
- **Disadvantages:** Costlier and bulkier than twisted pair; needs proper grounding.
- **Applications:** Cable TV (CATV), broadband Internet and CCTV.

3. Optical Fiber Cable

Sends data as light through a glass/plastic **core** and **cladding** using **total internal reflection**. Very high speed, long distance, low loss; WDM carries many wavelengths on one fibre.

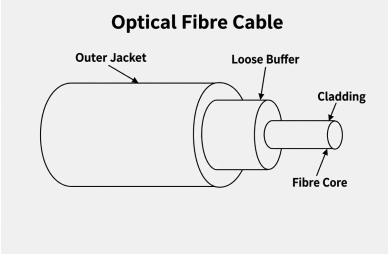

Figure: Structure of optical fibre cable

- **Advantages:** Very high bandwidth; low attenuation; immune to EMI; lightweight; resistant to corrosion.
- **Disadvantages:** High cost; fragile; needs skilled splicing and careful handling.
- **Applications:** Internet backbones, undersea cables, telecom links.

Q30. Compare Twisted Pair Cable, Coaxial Cable, and Optical Fiber Cable based on structure, advantages, disadvantages, and applications. L4 · 5M

ANSWER

Twisted pair, coaxial cable and optical fibre are the three main guided media. They differ in construction, signal type, performance and use.

Point	Twisted Pair	Coaxial	Optical Fiber
Structure	Insulated copper pairs twisted together; UTP (no shield) or STP (foil/braid shield)	Copper core + dielectric insulator + metallic braid/foil shield + outer jacket	Glass/plastic core + cladding + buffer + outer jacket
Signal type	Electrical	Electrical (baseband or broadband)	Light (total internal reflection)
Bandwidth	Moderate	Higher than twisted pair	Very high (supports WDM)
Distance / attenuation	Short to medium; higher attenuation	Medium; better than twisted pair	Very long; very low attenuation
EMI / noise	More affected (UTP); STP is better	Good shield; less EMI	Immune to EMI and electrical noise
Advantages	Lowest cost; light; easy to install and maintain	Higher bandwidth; better shielding; supports FDM/multi-channel	Highest speed/capacity; long distance; secure; light; no corrosion
Disadvantages	Limited distance; more crosstalk/noise; lower capacity than others	Bulkier; costlier than TP; needs grounding; can be tapped	High cost; fragile; needs skilled splicing and special equipment
Applications	LANs (Ethernet), telephone lines; STP in industrial/noisy areas	Cable TV (CATV), broadband Internet, CCTV	Internet backbones, undersea cables, telecom, long-haul links

Conclusion: Use **twisted pair** for low-cost short-distance LANs, **coaxial cable** for TV/broadband links needing better shielding, and **optical fibre** when very high bandwidth and long-distance communication are required.

UNGUIDED TRANSMISSION MEDIA

Key Terms

Term	Definition
Unguided media	Wireless (unbounded) media where signals travel through air or free space without a physical cable.
Radio waves	Electromagnetic waves that can travel in all directions and often penetrate buildings; used for broadcast and mobile links.
Microwaves	High-frequency waves that usually need line-of-sight (LOS) between antennas; used in cellular, satellite and TV links.
Infrared (IR)	Very short-range optical wireless signals that cannot pass through walls; used in remotes and nearby peripherals.
Line of sight (LOS)	A clear straight path between transmitter and receiver antennas (needed for most microwave links).

2 Marks

Q31. What is Unguided Transmission Media? List its types. **L1 · 2M**

ANSWER

Unguided transmission media (also called wireless or unbounded media) carry signals through air or free space using electromagnetic waves. No physical cable is required between sender and receiver.

Types:

1. Radio Waves
2. Microwaves
3. Infrared

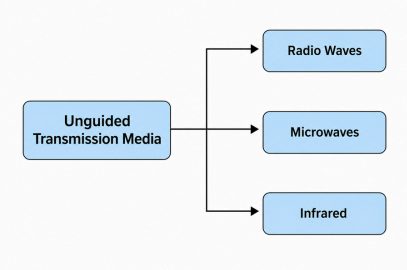

Figure: Types of unguided transmission media

Q32. Distinguish between Radio Waves and Microwaves. **L2 · 2M**

ANSWER

Point	Radio Waves	Microwaves
Direction	Omnidirectional (all directions)	Highly directional (point-to-point)
Line of sight	Not always required; can pass through buildings	Usually requires clear LOS between antennas
Frequency	About 3 kHz to 300 GHz (lower bands common for broadcast)	About 1 GHz to 300 GHz
Applications	AM/FM radio, TV, cordless phones, mobiles	Cellular backhaul, satellite, TV distribution

5 Marks

Q33. Explain the different types of unguided transmission media with neat diagrams. L2 · 5M

ANSWER

Unguided media transmit data as electromagnetic waves through air or free space (no cable). The three main types are Radio Waves, Microwaves and Infrared.

1. Radio Waves

Radio waves are easy to generate and can travel in all directions (**omnidirectional**). They can penetrate buildings, so strict line of sight is not always needed. They are widely used for broadcasting and wireless communication.

- **Frequency:** about 3 kHz to 300 GHz
- **Applications:** AM/FM radio, television, cordless phones, mobile communication

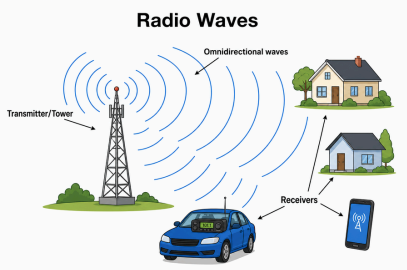

Figure: Radio waves (omnidirectional transmission)

2. Microwaves

Microwaves are high-frequency waves used mainly for **line-of-sight** links. Transmitting and receiving antennas must be properly aligned. Coverage depends on antenna height. They support high data rates over long distances without laying cables.

- **Frequency:** about 1 GHz to 300 GHz
- **Advantages:** No land cables needed; good for difficult terrain and oceans; high data rate
- **Disadvantages:** Needs LOS; affected by rain/fog; antenna cost; less secure without encryption
- **Applications:** Mobile/cellular links, satellite communication, TV distribution

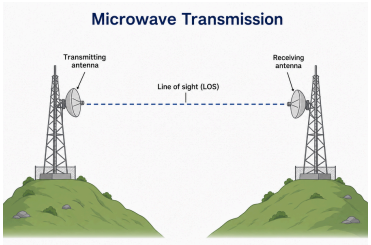

Figure: Microwave transmission (line-of-sight antennas)

3. Infrared

Infrared waves are used for **short-range** wireless links. They cannot penetrate solid obstacles such as walls, so interference between nearby rooms is low, but range is limited. Line of sight is usually required between devices.

- **Frequency:** about 300 GHz to 400 THz
- **Applications:** TV remotes, wireless keyboards, mice and short-range peripherals
- **Limitation:** Not suitable for long outdoor links because of high attenuation and obstacles

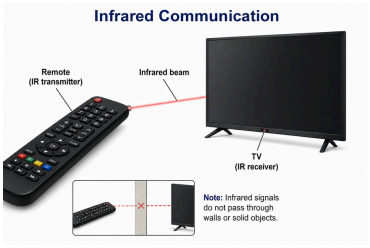

Figure: Infrared communication (remote to TV)

Point	Radio	Microwave	Infrared
Direction	Omnidirectional	Directional (LOS)	Directional (LOS)
Obstacles	Can penetrate buildings	Poor penetration	Cannot penetrate walls
Range	Long	Long (with towers/satellite)	Very short